

RehabOS — IASF 2027 Design Tool Content Pack

Sep 24, 2026 · @Aditya

Ready-to-paste content for the four IUCEE IASF 2027 workbooks, written against what RehabOS actually does.

How to use this pack

Each table below is one sheet's entry rows, in the sheet's own column order — paste a block, then fill the columns the workbook calculates for you. Wide sheets are split into two tables keyed by ID so they stay readable; paste the first block into columns A onwards, the second into the columns named in its header.

This section	Workbook	Sheet	Entry rows available
Tool 1 — Requirements	Requirements_Traceability_Matrix.xlsm	Traceability Matrix	rows 7–26 (20)
Tool 2 — Verification	Verification_Evidence_Matrix.xlsm	Verification Evidence	rows 7–26 (20)
Tool 3 — Validation	Validation_Evidence_Matrix.xlsm	Validation Evidence	rows 7–26 (20)
Tool 4 — DFMEA	IASF_S4_Tool_DFMEA.xlsx	DFMEA_Template	rows 14–21 (8)

Never type into these — they are formulas. Traceability Matrix column L (Traceability check) and DFMEA columns J, K and S (RPN, AP, RPN After) fill themselves. The DFMEA sheet only accepts 1–10 in columns E, G and I.

ID scheme used throughout. REQ-0NN for requirements, T-0NN for test cases, EV-0NN for evidence, DES-0NN when one requirement splits into several design statements. Every test case ID and evidence ID in Tools 2 and 3 matches the requirement row that owns it, so the traceability check reads Complete.

Evidence locations point at files in this repository (test suites, verification scripts, screenshots). A reviewer can open any of them, which is what the evidence-quality checklist in the workbooks asks for.

Status honesty. Rows marked Pass have a result already measured. Rows marked Pending or Did not Test are planned but unrun — the last section lists them so nothing is claimed that has not happened.

What the four workbooks are

They are the IUCEE IASF 2027 Session 4 design-assurance toolset, and they are meant to be filled as one chain: a need becomes a measurable requirement (Tool 1), which is proved built-right (Tool 2) and right-thing-built (Tool 3), while DFMEA (Tool 4) attacks the same design for ways it could fail. All four ship with a worked example from a different project, "Stride to Sustain", a piezoelectric stair-tile energy harvester — the example rows are teaching material, not a template to imitate literally.

Workbook	Sheets	Columns	What it proves
Requirements_Traceability_Matrix.xlsm	How to Use, Traceability Matrix, Example	A-M (13)	Every stakeholder need reaches evidence, and back again
Verification_Evidence_Matrix.xlsm	How to Use, Verification Evidence, Example	A-O (15)	The product meets its written specification
Validation_Evidence_Matrix.xlsm	How to Use, Validation Evidence, Example	A-O (15)	The product solves the real problem in real conditions
IASF_S4_Tool_DFMEA.xlsx	How_to_Use, DFMEA_Template, Piezo_Example	A-S (19)	Failure modes are ranked and the risky ones have owned actions

The dropdown values the sheets enforce

Typing anything else fails validation, so the tables in this pack only use these.

Field	Sheet	Allowed values
Priority (E)	Traceability	Must, Should, Could
Requirement type (F)	Traceability	Performance, Safety, Reliability, Physical, Usability, Durability, Human factors, Interface, Regulatory
Validation method (H / D)	Traceability, Validation	Measurement, Instrumented test, Runtime test, Charging test, Thermal test, Drop test, Dimensional inspection, User validation, Cycle test, Analysis, Inspection, Demonstration
Method (D)	Verification	Measurement, Instrumented test, Simulation, Bench Test, Hand Calculation, Pilot test with users, Inspection review with experts
Validation status (K)	Traceability	Pass, Fail, Not Run, Pending, Blocked
Result (M)	Validation	Pass, Fail, Not Run, Blocked
Result (M)	Verification	Pass, Fail, Did not Test, Blocked - Unable to test

How DFMEA scores are judged

Severity is the effect on the patient and does not soften because a failure is rare: 9–10 is safety or regulatory harm, 7–8 product inoperable, 5–6 reduced performance, 3–4 minor, 1–2 none. Occurrence runs 9–10 for daily down to 1–2 for once in years. Detection is inverted — 1–2 means the defect is always caught, 9–10 means it reaches the patient unseen. RPN is $S \times O \times D$, and the sheet marks Action Priority H when $S \geq 9$ or $RPN \geq 200$, M at $RPN \geq 100$, L below. Every H row needs an action before design release.

Two things to fix in the folder

1. **The four workbooks are duplicated into `app/src/./public/models/`.** The copies are byte-identical to the ones in the project root and are not tracked by git. Anything under `public/` is copied verbatim into the production build, so the next `npm run build` publishes your DFMEA and evidence matrices at URLs like `/models/IASF_S4_Tool_DFMEA.xlsx`, downloadable by anyone with the site address. Delete the four copies under `public/models/` and keep the root ones.

2. The Verification workbook's Purpose cell is truncated. It begins mid-sentence: "When that happens it is not a testing failure — it is a requirements failure, and it points back to Session 2." The opening lines that explain what verification is are missing, unlike the matching cell in the Validation workbook. Worth asking your mentor for the intact file before you rely on that sheet's instructions.

Tool 1 — Requirements Traceability Matrix

Sixteen requirements covering the rehabilitation gap RehabOS attacks: a patient who cannot travel to a clinic must still get correct exercise, correct dose, and a record their physiotherapist can trust. Paste this first block into columns A–G of the Traceability Matrix sheet, starting at row 7.

A Requirement ID	B Stakeholder / Source	C Customer need	D Measurable requirement statement	E Priority	F Requirement type	G Allocated design feature / subsystem
REQ-001	Patient interview; physiotherapist consultation	Patients need to know their exercises counted, without anyone watching them	The system shall count completed repetitions of each prescribed exercise with at least 95% agreement against a human observer, for a patient 1.5–2.5 m from a 720p webcam in room lighting of 100 lux or more	Must	Performance	Pose engine (PoseDetectorTasksVision) + repetition state machine (GenFeedback, rehabFactory)
REQ-002	Patient interview	Correction must arrive while the patient is still in the movement, not afterwards	The system shall display and speak form feedback within 500 ms of the pose frame that triggered it, on the reference low-end device	Must	Performance	GenFeedback state transitions; shared voice queue (coachSpeech)

A Requirement ID	B Stakeholder / Source	C Customer need	D Measurable requirement statement	E Priority	F Requirement type	G Allocated design feature / subsystem
REQ-003	Data-protection review; patient concern about being filmed at home	Patients will not use a camera at home if video could leave the house	No camera frame, video segment or derived image shall leave the patient device or be written to persistent storage; only numeric landmarks and scores shall be retained	Must	Regulatory	In-browser MediaPipe pipeline (pose and face landmarker); MoodMirror capture loop
REQ-004	Physiotherapist consultation	Patients copy what they can see; text instructions are not enough	The 3D coach shall demonstrate each of the prescribed rehabilitation exercises with an anatomically correct, looping motion, covering 100% of exercises available in the catalogue	Must	Interface	AvatarScene, AvatarRig, exerciseMotions (31 motions)
REQ-005	Field observation of a live session	The coach must not vanish or freeze part-way through a session	The 3D coach shall complete a 20-minute exercise session with zero unrecovered WebGL context losses, and the number of live WebGL contexts shall not grow during the session	Must	Reliability	AvatarScene renderer lifecycle; AvatarCoach WebGL support probe
REQ-006	Hindi-speaking patient feedback	Spoken guidance must be a whole, understandable sentence	Every spoken cue shall be delivered without interruption by a later cue or by exercise feedback, with zero mid-sentence cancellations measured over a full exercise cycle	Must	Usability	coachSpeech shared voice queue; Audio.playText

A Requirement ID	B Stakeholder / Source	C Customer need	D Measurable requirement statement	E Priority	F Requirement type	G Allocated design feature / subsystem
REQ-007	Hindi-speaking patient feedback; mentor review	Elderly patients in India need instructions in their own language, including angles	The coach shall present 100% of on-screen and spoken cues in Hindi when Hindi is selected, with numeric targets and the degree symbol spoken as Hindi words	Must	Human factors	coachI18n (EN/Hi strings); speechText normaliser
REQ-008	Physiotherapist consultation	Therapists need to know whether the prescribed plan was actually done	The system shall record completion of each prescribed plan item per calendar day and report a consecutive-day streak that resets after one missed day	Should	Performance	today'sPlan and planDone (patientData); PatientDashboard
REQ-009	Physiotherapist consultation	Therapists need objective proof that joint range is improving	The system shall record minimum and maximum joint angle per exercise session and report the change in range over the last 14 days, rejecting frame-to-frame jumps greater than 40 degrees as tracking noise	Must	Performance	ExercisePage recordRange; mobilityByExercise, mobilityChange
REQ-010	Patient interview; mentor review	Daily mood entry is a chore, and low mood predicts drop- out	The system shall derive a daily mood on a 1–5 scale from a five-second camera check- in, using facial expression signals, and shall award check-in points independently of the mood recorded	Should	Usability	faceMood scoring, faceMoodDetector, MoodMirror, HappyMeterCard

A Requirement ID	B Stakeholder / Source	C Customer need	D Measurable requirement statement	E Priority	F Requirement type	G Allocated design feature / subsystem
REQ-011	Clinical safety review	A mistyped vital sign could mislead the clinician	The system shall reject any manually entered vital outside its clinical plausibility range before saving, covering heart rate 30– 220 bpm, systolic 70–250 mmHg, diastolic 40–150 mmHg, SpO2 70–100%, temperature 34– 42 C and sleep 0–24 h	Must	Safety	HealthLogDialog field ranges; addHealthLog
REQ-012	Elderly user observation	Small text and small buttons exclude the people this product is for	Every interactive control on the patient journey shall present a touch target of at least 44 x 44 px and body text of at least 16 px, at a contrast ratio of 4.5:1 or better in both light and dark themes	Must	Human factors	MUI theme typography and spacing; patient dashboard components
REQ-013	Primary user need (rehabilitation gap)	A patient with limited mobility must manage a session alone at home	At least 80% of first-time patients aged 65 and over shall complete one full prescribed exercise session unaided, with no more than one prompt from an observer	Must	Usability	End-to-end patient journey: dashboard, plan, exercise page, coach

A Requirement ID	B Stakeholder / Source	C Customer need	D Measurable requirement statement	E Priority	F Requirement type	G Allocated design feature / subsystem
REQ-014	Physiotherapist consultation	Remote review must be possible when travel is not	The system shall let a patient request and hold a video session with a physiotherapist, with the appointment state visible to both parties within 24 hours of the request	Should	Interface	BookSession, SessionRoom, TherapistDashboard, appointments store
REQ-015	Data-protection review	One patient must never see another patient's record	Health records shall be readable only by the authenticated account that owns them, and demo-mode activity shall never be written to the clinical datastore	Must	Regulatory	Firebase Auth; per-user Firestore document; demoPatient local-only store
REQ-016	Elderly user observation; low-cost device survey	Many patients own an older phone or laptop with weak graphics	The patient exercise screen shall sustain at least 20 pose frames per second with the 3D coach visible on the reference low-end device, without thermal shutdown over 20 minutes	Should	Reliability	Pose model (lite); coach frame-rate cap and pixel- ratio cap in AvatarScene

Paste this second block into columns H–K and M. Column L calculates itself and reads Complete only when B, C, D, G, H, I and J are all filled, which every row below satisfies.

A Requirement ID	H Validation method	I Test case ID	J Evidence ID	K Validation status	M Change-impact notes
REQ-001	Instrumented test	T-001	EV-001	Pending	Any change to joint thresholds in an exercise module re-opens T-001 and REQ-009
REQ-002	Measurement	T-002	EV-002	Pending	Voice queue changes affect perceived latency; retest with REQ-006

A Requirement ID	H Validation method	I Test case ID	J Evidence ID	K Validation status	M Change-impact notes
REQ-003	Inspection	T-003	EV-003	Pending	Adding any server-side analysis re-opens this row and REQ-015
REQ-004	Inspection	T-004	EV-004	Pass	Adding a catalogue exercise without a coach motion breaks this row
REQ-005	Instrumented test	T-005	EV-005	Pass	Any new WebGL surface on the exercise page re-opens T-005 and REQ-016
REQ-006	Instrumented test	T-006	EV-006	Pass	Cue timing in exerciseMotions changes the queue load; retest with REQ-002
REQ-007	Inspection	T-007	EV-007	Pass	Every new cue string needs a Hindi entry before release; affects REQ-006
REQ-008	Runtime test	T-008	EV-008	Pass	Changing plan identifiers invalidates stored completion keys
REQ-009	Analysis	T-009	EV-009	Pass	Tracking-noise threshold couples to REQ-001 accuracy
REQ-010	Instrumented test	T-010	EV-010	Pass	Rescoring expressions shifts historical mood trends; affects REQ-008 streaks
REQ-011	Inspection	T-011	EV-011	Pass	Adding a vital sign requires a new range and a retest
REQ-012	User validation	T-012	EV-012	Pending	Theme changes re-open this row; couples to REQ-013
REQ-013	User validation	T-013	EV-013	Pending	Depends on REQ-004, REQ-007 and REQ-012 all passing first
REQ-014	Runtime test	T-014	EV-014	Pending	Appointment schema changes affect the therapist dashboard
REQ-015	Inspection	T-015	EV-015	Pass	Any new write path must be checked against demo mode; affects REQ-003
REQ-016	Measurement	T-016	EV-016	Pending	Raising coach quality settings re-opens this row and REQ-005

Tool 2 — Verification Evidence Matrix

Verification asks whether the product meets its written specification, so these rows are instrumented, repeatable and mostly already run. Eight of the twelve carry a real measured result from this codebase; the rest are defined but unrun and are marked Did not Test rather than dressed up as passes. Paste the first block into columns A–H, the second into I–O.

A Req ID	B Test case ID	C Key question	D Method	E Procedure summary	F Sample size	G Test conditions	H Acceptance criteria
REQ-001	T-001	Does the repetition counter agree with a human counting the same movement?	Instrumented test	Record video of each rehab exercise, play it into the pose pipeline, compare machine count with two independent human counts of the same recording	10 sessions across 5 exercises	720p webcam, 1.5–2.5 m, 100–300 lux, plain background	Machine count within 5% of human count on every session, with no session off by more than one repetition
REQ-001	T-001b	Does the repetition state machine advance only on a genuine full-range movement?	Bench Test	Drive each exercise module with synthetic joint-angle sequences, including partial range and static hold, and assert the state transitions and counts	11 exercise modules	Automated test run, no camera	Every module counts full-range movement, ignores partial range, and holds do not increment the counter
REQ-002	T-002	How long after the triggering frame does feedback reach the patient?	Measurement	Timestamp the pose frame that causes a state change and the moment the cue is painted and spoken; take the 95th percentile over a session	200 transitions	Reference low-end device, coach visible, voice on	95th percentile under 500 ms
REQ-003	T-003	Does any image data leave the device?	Inspection review with experts	Capture all network traffic for a full session including a mood check-in; inspect every request body and all persisted storage for image or frame payloads	1 full session	Browser developer tools network capture; storage inspection	Zero requests carrying image, video or frame data; zero image bytes in local or remote storage

A Req ID	B Test case ID	C Key question	D Method	E Procedure summary	F Sample size	G Test conditions	H Acceptance criteria
REQ-004	T-004	Does every catalogue exercise have a correct coach demonstration?	Inspection review with experts	Load each exercise, record the coach loop, and have a physiotherapist mark the motion as anatomically acceptable or not	31 motions	Desktop browser, both coach characters	100% of catalogue exercises demonstrate, with no motion rejected by the reviewing physiotherapist
REQ-005	T-005	Do live WebGL contexts grow while a session runs?	Instrumented test	Patch canvas context creation to count contexts, open the exercise page in a headless browser, hold the session and sample the count at load and after a further 40 seconds	2 exercises, 55 s each	Headless Chrome, software rendering, fake camera feed	Context count does not increase between the two samples and no context-loss event is recorded
REQ-006	T-006	Is any spoken cue cut off by the next one?	Instrumented test	Replace the speech engine with a stub timed at 90 ms per character, run a full exercise cycle in Hindi and in English, and log every speak, end, cancel and overlap	2 languages, 1 full cycle each	Headless Chrome, coach voice on, lunge exercise	Zero cancellations while speaking and zero overlapping utterances in both languages
REQ-007	T-007	Are the Hindi cues complete, and is the angle spoken as a word?	Inspection review with experts	Compare the Hindi and English cue tables key by key; run the speech normaliser over every cue containing a degree symbol and inspect the spoken string	52 cue keys	Automated comparison plus unit tests	No missing Hindi key, and every numeric angle is spoken in Hindi words followed by the word for degrees

A Req ID	B Test case ID	C Key question	D Method	E Procedure summary	F Sample size	G Test conditions	H Acceptance criteria
REQ-009	T-009	Is the recorded joint range arithmetic correct and noise-resistant?	Bench Test	Feed known angle series, including a 60-degree single-frame jump, into the range recorder and the trend helpers, and assert the stored minimum, maximum and 14-day change	4 series	Automated test run	Minimum and maximum match the known series, the injected jump is discarded, and the reported change matches hand calculation
REQ-010	T-010	Does the expression scorer put real faces on the right part of the 1-5 scale?	Instrumented test	Run the face landmarker over labelled photographs in a browser, capture the expression signals, and score them with the production scoring function	4 photographs, all smiling	Headless browser, still images, image mode	Every smiling face scores 4 or higher, and a neutral signal set scores 3
REQ-011	T-011	Are out-of-range vitals rejected before saving?	Bench Test	Enter one value below, inside and above each field's range, and attempt to save each time	6 fields x 3 values	Patient health log dialog	Save is blocked for every out-of-range value and permitted for every in-range value
REQ-015	T-015	Can demo activity reach the clinical datastore, or one account read another's record?	Inspection review with experts	Exercise every write path in demo mode with the datastore observed; review authentication rules and the per-account document path	All write paths	Automated tests plus rules review	Zero demo writes reach the datastore and no rule permits cross-account reads

A Req ID	B Test case ID	I Actual result	J Evidence ID / location	K Owner	L Test date	M Result	N Corrective action / retest	O Instructor challenge prompt
REQ-001	T-001	Not yet run	EV-001 / (planned) accuracy study workbook	[Name]		Did not Test	Run before design release; blocks REQ-001 sign-off	Would a patient with a walking frame in the frame change the count?
REQ-001	T-001b	11 of 11 exercise modules pass their state-machine tests	EV-001b / app/src/utis/exercises/rehabExercises.test.js	[Name]	2026-09-24	Pass	None	Synthetic angles are ideal; what does a real tremor do to these transitions?
REQ-002	T-002	Not yet run	EV-002 / (planned) latency log	[Name]		Did not Test	Instrument the transition path, then measure	Is 500 ms fast enough to stop a patient completing a harmful repetition?

A Req ID	B Test case ID	I Actual result	J Evidence ID / location	K Owner	L Test date	M Result	N Corrective action / retest	O Instructor challenge prompt
REQ-003	T-003	Not yet run	EV-003 / (planned) network capture	[Name]		Did not Test	Run a full capture including a mood check-in	The model files load from a public CDN; does that request reveal anything about the patient?
REQ-004	T-004	31 coach motions present and demonstrating; physiotherapist sign-off outstanding	EV-004 / app/src/utlis/avatar/exerciseMotions.js plus session screenshots	[Name]	2026-09-24	Pass	Obtain the physiotherapist's written review before release	A motion that looks right on a young avatar may be unsafe for a 72-year-old; who checked that?
REQ-005	T-005	Contexts at load 15, after 40 s 39 before the fix; 5 and 5 after the fix, zero loss events	EV-005 / context count harness output, before and after	[Name]	2026-09-24	Pass	Fixed: the support probe now runs once and releases its context; coach panel no longer re-renders per frame	Software rendering was used; would a real weak GPU still drop the context?
REQ-006	T-006	11 utterances Hindi and 10 English per cycle, 0 cancellations while speaking, 0 overlaps	EV-006 / speech trace harness output	[Name]	2026-09-24	Pass	None	Cues now queue; does a patient ever hear guidance that is already out of date?
REQ-007	T-007	All 52 cue keys present in Hindi; the 90-degree cue speaks as the Hindi words for ninety degrees	EV-007 / app/src/utlis/avatar/coachI18n.js, coachSpeech.test.js, speech trace	[Name]	2026-09-24	Pass	None	Translation exists, but was it read aloud to a Hindi speaker for naturalness?
REQ-009	T-009	Range arithmetic and 14-day change match hand calculation; 60-degree jump discarded	EV-009 / app/src/utlis/patient/patientData.test.js, ExercisePage recordRange	[Name]	2026-09-24	Pass	None	If the camera loses the leg for a second, is the maximum still trustworthy?
REQ-010	T-010	All 4 photographs scored 4 or 5; neutral signal set scored 3	EV-010 / app/src/utlis/mood/faceMood.test.js real-face fixtures	[Name]	2026-09-24	Pass	Extend the sample with sad and neutral faces before relying on the low end of the scale	Every face in the sample was smiling; what has actually been proved?
REQ-011	T-011	Every out-of-range value blocked; in-range values saved	EV-011 / app/src/components/patient/HealthLogDialog.js field ranges	[Name]	2026-09-24	Pass	None	A plausible number can still be clinically wrong; what catches that?
REQ-015	T-015	Demo writes stay in the browser; records are keyed per account	EV-015 / app/src/utlis/patient/demoPatient.test.js, patientData write paths	[Name]	2026-09-24	Pass	Datastore security rules still need an expert review	Tests prove the client behaves; what stops a request made outside the app?

Tool 3 — Validation Evidence Matrix

Validation asks whether an elderly or mobility-limited patient is genuinely better served at home, so almost every row here needs real people and is therefore still unrun. That is the honest state of the project: the engineering is evidenced, the human evidence is not yet collected. Two rows can be marked Pass today because they were run against real recorded material rather than a live cohort.

A Req ID	B Test case ID	C Validation question	D Method	E Procedure summary	F Sample size	G Test conditions	H Acceptance criteria
REQ-013	T-013	Can a patient over 65 complete a prescribed session alone at home?	User validation	Give each participant only the device and their prescribed plan; an observer watches without helping and records every prompt needed, plus where the participant hesitated	10 participants aged 65+	Participant's own home, own device, own internet, no family assistance	At least 8 of 10 complete the full session with no more than one prompt
REQ-012	T-012	Can the patient read the screen and hit the controls without help?	User validation	Ask each participant to find today's plan, start an exercise and log a vital; record misses, mis-taps and reading difficulty; follow with a short interview	10 participants aged 65+	Normal home lighting, participant's usual glasses, light and dark theme	No participant fails to complete a task because of text size, contrast or target size
REQ-007	T-007v	Does a Hindi-speaking patient understand the spoken cue and act on it correctly?	User validation	Play each Hindi cue to the participant without showing the screen, and ask them to perform what they heard; a physiotherapist marks the movement as correct or not	8 Hindi-speaking participants, 12 cues each	Quiet room, device speaker at normal volume	At least 90% of cues are performed correctly, with no cue misunderstood by more than one participant
REQ-001	T-001v	Does the patient trust the count enough to rely on it?	User validation	After a supervised session, show the patient the recorded count and ask whether it matches what they felt they did; record disagreements and their cause	10 patients	Live home session with a physiotherapist observing remotely	No patient reports a count they believe is wrong by more than one repetition
REQ-004	T-004v	Does watching the 3D coach produce safer movement than written instructions?	Demonstration	Two groups perform the same unfamiliar exercise, one with the coach visible and one with text and a still image; a blinded physiotherapist scores form from recordings	2 groups of 8	First exposure to the exercise, no prior coaching	The coach group scores higher on the physiotherapist's form rubric, with no participant in either group performing an unsafe movement
REQ-009	T-009v	Does the range-of-motion trend tell the physiotherapist something they can act on?	Inspection	Give three physiotherapists two weeks of a real patient's recorded data and ask what they would change in the plan, then compare against their judgement from a normal clinic review	3 physiotherapists, 5 patient records	Blinded to which patient is which	All three reach a clinical decision from the recorded data alone, and none contradicts their own clinic-based judgement

A Req ID	B Test case ID	C Validation question	D Method	E Procedure summary	F Sample size	G Test conditions	H Acceptance criteria
REQ-010	T-010v	Do patients accept a camera mood check-in, and does it reflect how they actually feel?	User validation	Patient completes the camera check-in, then independently reports their mood on the same 1-5 scale without seeing the result; compare the pair and interview on comfort	15 patients, 7 days each	Home use, once daily, unsupervised	Agreement within one scale point on at least 70% of check-ins, and no participant withdraws because of discomfort with the camera
REQ-010	T-010b	Does the scoring hold up across faces it has not seen?	Measurement	Score a labelled public expression set with the production scoring function, covering happy, neutral and sad faces	60 images, balanced across 3 classes	Still images, on-device scoring	Happy scores above neutral and neutral above sad in at least 80% of pairwise comparisons
REQ-008	T-008v	Does the plan and streak actually change whether people keep exercising?	Runtime test	Compare completion rates over four weeks between patients using the plan with streaks and a group using the same plan on paper	2 groups of 12, 4 weeks	Unsupervised home use	Adherence in the app group is at least 15 percentage points higher
REQ-014	T-014v	Can a patient hold a useful remote session on a weak connection?	Runtime test	Run a scheduled physiotherapy session end to end on a throttled mobile connection; the physiotherapist records whether they could assess and advise	6 sessions	3G-class connection, 1 Mbps up, patient on a phone	All 6 sessions complete, and the physiotherapist reports the assessment was possible in at least 5
REQ-005	T-005v	Does the coach hold up on the hardware patients really own?	Runtime test	Run a 20-minute session on the three most common low-end devices in the target group and record any coach interruption visible to the patient	3 device models, 3 runs each	Patient-representative devices, battery power, normal room temperature	No visible coach interruption in any run
REQ-003	T-003v	Do patients believe their privacy is protected, and is the claim true to them?	Demonstration	Show the patient where processing happens, then ask them to state in their own words what leaves their device; record misconceptions	10 patients	After first use of both camera features	At least 8 of 10 correctly state that no video leaves the device, and none reports being deterred from using the camera

A Req ID	B Test case ID	I Actual result	J Evidence ID / location	K Owner	L Test date	M Result	N Corrective action / retest	O Instructor challenge prompt
REQ-013	T-013	Not yet run	EV-013 / (planned) observation notes and session recordings	[Name]		Not Run	Recruit through the partner clinic; this is the project's central claim and is currently unevidenced	The whole product exists to remove the clinic visit; what proves it did?
REQ-012	T-012	Not yet run	EV-012 / (planned) usability session recordings	[Name]		Not Run	Run alongside T-013 with the same participants	Who on the team has watched a 72-year-old use this unaided?
REQ-007	T-007v	Not yet run	EV-007v / (planned) cue comprehension sheets	[Name]		Not Run	Needs a Hindi-speaking physiotherapist to score the movements	The translation is literal and correct; is it how a physiotherapist would actually speak to a patient?
REQ-001	T-001v	Not yet run	EV-001v / (planned) post-session interview notes	[Name]		Not Run	Run after T-001 establishes machine accuracy	Trust and accuracy are different things; which one drives drop-out?
REQ-004	T-004v	Not yet run	EV-004v / (planned) blinded form-scoring sheets	[Name]		Not Run	Needs a physiotherapist willing to score blind	If the coach makes no measurable difference to form, what is the 3D work buying?
REQ-009	T-009v	Not yet run	EV-009v / (planned) clinician decision forms	[Name]		Not Run	Use the demo patient record as the rehearsal case, then real records	Would a physiotherapist change a plan on this data, or ask the patient to come in anyway?

A Req ID	B Test case ID	I Actual result	J Evidence ID / location	K Owner	L Test date	M Result	N Corrective action / retest	O Instructor challenge prompt
REQ-010	T-010v	Not yet run	EV-010v / (planned) paired mood log	[Name]		Not Run	Depends on T-010b first; do not field the feature on a 4-image sample	A patient can smile on cue; what is the check-in actually measuring?
REQ-010	T-010b	Not yet run	EV-010b / (planned) expression dataset results	[Name]		Not Run	Highest-value quick test in this table; run before any patient sees a mood score	The current fixtures contain no sad face at all
REQ-008	T-008v	Not yet run	EV-008v / (planned) adherence comparison	[Name]		Not Run	Needs four weeks; start early or drop the claim	Streaks can shame a patient who is unwell; was that considered?
REQ-014	T-014v	Not yet run	EV-014v / (planned) session logs	[Name]		Not Run	Throttled-network rehearsal can be run in a lab this week	Patients who cannot travel often have poor connectivity too
REQ-005	T-005v	Not yet run on physical devices	EV-005v / (planned) device test log	[Name]		Not Run	Headless evidence exists (T-005); physical devices still required	Software rendering proved the leak was gone, not that the GPU copes
REQ-003	T-003v	Not yet run	EV-003v / (planned) interview notes	[Name]		Not Run	Pair with the network capture in T-003 so the claim and the belief are both evidenced	A true privacy claim that patients do not believe still loses the patient

Tool 4 — DFMEA

Eight failure modes, filling the template's eight rows. Two of them are not hypothetical: the coach renderer and the voice subsystem both failed in real use during development, and both now carry a completed action with re-rated scores, which is exactly what the tool's Step 13 asks for. The other six are live risks with actions still to take.

Header block (rows 4–9)

Field	Value
Project / Product Name	RehabOS — camera-guided home rehabilitation platform
Team Name	[your team name]
Date	[submission date]
Mentor	[mentor name]
Institution	[your institution]
Stage	Stage 3 — Concept Design / POC (change to Stage 4 if your programme counts a working prototype)

Columns A–D

A Component / Element	B Function & Performance Target	C Potential Failure Mode	D Potential Effect of Failure
Pose tracking module (MediaPipe pose landmarker, lite)	Track 33 body landmarks at 20 fps or more for a patient 1.5–2.5 m from the camera	Limb landmarks are lost or misplaced in dim light, cluttered rooms, or when a walking frame or chair occludes the leg	Repetitions are counted that did not happen, or real ones are missed; the patient exercises beyond or below the prescribed dose
Repetition state machine (GenFeedback, rehabFactory)	Count a repetition only when the prescribed joint travels its full target range	A compensatory movement is accepted as a valid repetition, for example hip hiking instead of bending the knee	The therapist sees progress that did not occur, and the patient rehearses a movement pattern that can delay or damage recovery
3D coach renderer (AvatarScene, WebGL)	Show the coach performing the exercise for a whole session at 30 fps	The browser drops the WebGL context and the coach disappears into a repeating "restoring" loop	The patient loses the demonstration mid-exercise and either stops or continues unguided with no model to copy

A Component / Element	B Function & Performance Target	C Potential Failure Mode	D Potential Effect of Failure
Coach voice subsystem (coachSpeech, exercise feedback audio)	Speak each cue as one complete sentence in the selected language	A cue is cancelled part-way by the next cue or by form feedback, and the degree symbol is not spoken at all	The patient hears "lower until both knees are at" with no angle, and loads the joint to a guessed range
Camera data path (pose and face pipelines)	Process every frame on the patient's device and retain numbers only	A later feature, analytics or crash reporting carries a frame, image or landmark stream off the device	Biometric health data leaves the patient's home without consent; regulatory breach and loss of patient trust
Manual vitals entry (health log)	Capture home vitals within clinically plausible ranges	A plausible but wrong value is entered, for example systolic and diastolic swapped on a small phone keyboard	The clinician reads a false picture of stability and delays escalation for a patient who is deteriorating
Mood check-in (face expression scoring)	Derive a daily mood on a 1–5 scale from a five-second camera check-in	A polite or masked smile is scored as good mood, so a persistently low mood never appears in the record	Deterioration in a patient's mental state is not surfaced to the physiotherapist, and drop-out or self-harm risk is missed
Tele-session (booking and video room)	Connect patient and physiotherapist at the scheduled time	The session fails to connect on a weak home connection, or the request is never confirmed by a therapist	The patient waits for care that does not arrive, loses confidence in remote rehabilitation, and may not rebook

Columns E–I (J, K calculate themselves)

RPN and AP are shown so you can check the sheet agrees: $RPN = S \times O \times D$, and AP is H when S is 9 or more or RPN is 200 or more, M at 100 or more, otherwise L.

A Component	E S	F Potential Cause of Failure	G O	H Current Detection Controls	I D	(J RPN)	(K AP)
Pose tracking module	6	Lightweight model chosen for low-end devices; no minimum lighting or framing check before a session starts; mobility aids occlude the limb	6	Live skeleton overlay lets the patient see tracking quality; motion gate rejects frames with no movement; 11 automated exercise-module tests	4	144	M
Repetition state machine	7	Counting logic watches a single joint angle with no check on the rest of the body	5	Per-exercise angle thresholds; jumps over 40 degrees discarded as noise; automated state-machine tests	5	175	M
3D coach renderer	5	The WebGL support probe opened a new graphics context on every render, and the panel re-rendered on every tracked frame, exceeding the browser's context limit	7	Context-loss handler with a visible recovering overlay; context-count harness in a headless browser	3	105	M
Coach voice subsystem	7	Every cue cancelled the speech engine, and exercise feedback cancelled it independently; the degree symbol was passed to the engine unchanged	8	None before this analysis — the cues were only ever checked by ear	6	336	H
Camera data path	9	No automated guard prevents a future contributor from passing a camera surface into a network call	2	Code review only; no network capture in the release checklist	6	108	H
Manual vitals entry	8	Fields are validated one at a time, with no cross-field sanity check; elderly patients type on small keyboards	5	Per-field plausibility ranges; both blood-pressure numbers required together	4	160	M

A Component	E S	F Potential Cause of Failure	G O	H Current Detection Controls	I D	(J RPN)	(K AP)
Mood check-in	7	Scoring was tuned on four smiling faces with no low-mood samples; a single five-second window is easy to perform for	6	Manual face picker as fallback; a low score shows advice to contact a physiotherapist or doctor	6	252	H
Tele-session	6	No bandwidth adaptation or audio-only fallback; confirmation depends on a therapist acting manually	5	Appointment status visible on both the patient and therapist dashboards	4	120	M

Columns L-S (actions)

Leave the After columns blank until the action is actually done — the sheet's Step 13 is explicit about that. Only the two rows that were fixed during development carry re-rated scores.

A Component	L Recommended Action	M Responsible	N Target Date	O Action Taken	P S After	Q O After	R D After	(S RPN After)
Pose tracking module	Add a lighting and framing check before the session starts, and pause counting when landmark confidence drops below threshold rather than counting through it	[Name]	Before pilot					

A Component	L Recommended Action	M Responsible	N Target Date	O Action Taken	P S After	Q O After	R D After	(S RPN After)
Repetition state machine	Add a second joint guard per exercise, for example pelvis level during a straight leg raise, and flag suspected compensation to the therapist instead of counting silently	[Name]	Before pilot					
3D coach renderer	Run the WebGL support probe once and release its context; stop the coach panel re-rendering on every tracked frame; cap frame rate and pixel ratio; step quality down after repeated context loss	[Name]	Done	Probe now runs once and releases its context; coach panel memoised; 30 fps and 1.5x pixel-ratio caps added. Measured contexts over 55 s fell from 39 to 5, with zero loss events	5	2	3	30
Coach voice subsystem	Route the coach and exercise feedback through one speech queue so a sentence always finishes; rewrite symbols and numbers into spoken words before speaking; add an instrumented speech trace to the release checklist	[Name]	Done	Shared queue added, newest cue waits behind the current one; degree symbol and Hindi numerals spoken as words; traced at 0 cancellations and 0 overlaps over a full cycle in both languages	7	2	3	42

A Component	L Recommended Action	M Responsible	N Target Date	O Action Taken	P S After	Q O After	R D After	(S RPN After)
Camera data path	Add an automated check that fails the build if a camera surface reaches a network call, and add a network capture to the release checklist	[Name]	Before design release					
Manual vitals entry	Add cross-field validation, starting with systolic at least 20 mmHg above diastolic, and show the previous reading beside the input for comparison	[Name]	Before pilot					
Mood check-in	Validate the scoring on a balanced labelled expression set before any patient sees a mood score; escalate a sustained low mood combined with rising pain to the physiotherapist rather than treating mood as decoration	[Name]	Before design release					
Tele-session	Add an audio-only fallback, and remind the therapist automatically when a request is unconfirmed after 12 hours	[Name]	Before pilot					

If your mentor asks for more than eight rows, the next two to add are demo-mode data reaching a real patient record (S 7) and cross-account data exposure (S 9, which forces AP H).

Gaps to close before you submit

The engineering side is evidenced and the human side is not: 8 of 12 verification rows have a measured result, and 0 of 12 validation rows do. That asymmetry is the most honest thing in this pack, and a mentor will find it in thirty seconds, so it is better stated than hidden.

Run these three first — they are cheap and they change what you may claim.

1. **T-010b, the expression dataset.** The mood scale is currently tuned on four photographs, every one of them smiling. Until a balanced set is scored, the low end of that scale is an assumption, and REQ-010 should not be described as working.
2. **T-003, the network capture.** The privacy claim is the reason a patient lets a camera into their bedroom. It takes one session with the network tab recording to turn it from a design intention into evidence.
3. **T-005v, three real devices.** The context-loss fix was measured under software rendering, which cannot reproduce a real GPU running out of memory. Twenty minutes on three patient-representative phones settles it.

Then the human evidence. T-013 and T-012 with ten patients over 65 are the ones that speak to the rehabilitation gap itself. Everything else in this pack proves the software behaves; only those prove the product removes a clinic visit. If recruiting ten patients is not possible in your timeline, run four and say so in the sample-size column rather than leaving the row Not Run — a small honest sample with its limits stated scores better than an empty row.

Two points a mentor is likely to press, and the honest answers.

- *Why is the coach's after-action occurrence 2 and not 1?* Because the leak that caused the losses is fixed and measured, but a weak GPU under memory pressure can still drop a context; the degradation path exists precisely because that residual risk is real.
- *Why does a smile earn the same points as a low mood?* Because scoring mood would teach patients to perform a smile, and the physiotherapist needs the honest trend. The points reward checking in; the meter is what moves with how the patient feels.

Housekeeping. Delete the four workbook copies under `public/models/` before your next build, or they ship with the deployed site.